A Digitally Assisted Analog Front-End for Low-Cost Geophone Instrumentation

Elías D. Álvarez Martínez, Vicente González, *Member, IEEE,* and Enrique A. Vargas Cabral, *Member, IEEE*
Facultad de Ciencias y Tecnología, Universidad Católica "Nuestra Señora de la Asunción"
Asunción, Paraguay — {elias.alvarez, vgonzalez, evargas}@uc.edu.py

***Abstract*—Low-cost programmable mixed-signal platforms provide an attractive foundation for scientific instrumentation, but the limited precision of their configurable analog building blocks often constrains analog front-end performance. This paper proposes a Digitally Assisted Analog Front-End architecture that temporarily exploits the embedded mixed-signal resources of a programmable system-on-chip (PSoC) to improve the operating conditions of the analog signal chain while preserving a conventional acquisition path during normal operation.**

As a proof of architectural feasibility, the proposed architecture is demonstrated through an automatic foreground DC-offset calibration that sequentially recenters the programmable-gain amplifier, band-pass filter, summing stage, and low-pass filter using the on-chip ADC, DAC, and digital filtering resources.

The architecture was experimentally evaluated on two independently assembled geophone acquisition boards under eight board–gain configurations. Relative to a common fixed-reference baseline, the calibrated low-pass output reduced the residual DC offset from 2.9–45.8% to 0.4–3.9% of the nominal ADC input span using a single controller configuration across all reported conditions.

These results demonstrate the feasibility of the proposed Digitally Assisted Analog Front-End architecture as a practical approach for improving the operating conditions of low-cost configurable analog hardware using embedded mixed-signal resources. Statistical validation over larger hardware populations and broader environmental characterization remain future work.

***Index Terms*—Digitally assisted analog design, analog front end, mixed-signal systems, geophone instrumentation, programmable system-on-chip (PSoC).**

I. Introduction

Modern programmable mixed-signal platforms have become an attractive foundation for low-cost scientific instrumentation because they integrate configurable analog resources together with analog-to-digital and digital-to-analog converters, programmable digital filters, direct memory access, and embedded processors within a single device. However, the configurable analog building blocks available in these platforms generally provide lower precision, larger offsets, and reduced matching than dedicated analog integrated circuits, limiting the achievable performance of the analog front end.

Rather than addressing these limitations through higher-grade analog components, manual trimming, or additional external circuitry, the proposed work follows the principles of digitally assisted analog design introduced by Murmann and Boser [1] and further developed by Murmann [2]. In this work, a *Digitally Assisted Analog Front-End* denotes an architecture in which embedded digital resources temporarily assist the analog circuitry to improve its effective operating conditions while leaving the acquisition signal path unchanged during normal operation. Although digitally assisted analog design has primarily been investigated at the integrated-circuit level, the present work extends the same philosophy to low-cost scientific instrumentation through a geophone acquisition platform.

Low-cost multichannel geophone acquisition systems provide a representative application for this architecture. Surface-wave methods such as Multichannel Analysis of Surface Waves (MASW) estimate Rayleigh-wave dispersion to infer shear-wave velocity profiles for geotechnical site characterization [3], [4]. Low-cost embedded geophone interfaces have already been demonstrated [5], [6], but in multistage analog conditioning chains, accumulated amplifier and reference offsets progressively reduce the available dynamic range and, at high gain, may saturate intermediate stages before the microvolt-level seismic signal reaches the acquisition converter. Improving the effective operating conditions of the analog front end therefore directly benefits the overall measurement system.

The proposed front end forms part of a low-cost reconfigurable Internet-of-Things (IoT) geophone acquisition platform [7] based on SM-24 moving-coil geophones [8]. The PSoC 5LP integrates programmable analog blocks, voltage DACs (VDACs), a delta-sigma ADC, programmable digital filters, DMA resources, and an Arm Cortex-M3 processor [9], providing all resources required to implement the proposed architecture. As a proof of architectural feasibility, this work implements an automatic foreground DC-offset calibration that sequentially recenters the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) before acquisition.

Previous offset-compensation techniques generally follow three approaches: numerical estimation and digital correction [10], [11], physical trimming of individual analog stages [12], or autozero and chopper stabilization techniques [13], [14]. In contrast, the proposed architecture employs embedded digital resources only during foreground calibration. Once the desired operating points have been established, the calibration subsystem is disabled and the analog signal chain operates conventionally throughout data acquisition.

The contributions of this work are threefold: (i) the proposal of a Digitally Assisted Analog Front-End architecture that exploits embedded mixed-signal resources to improve the effective operating conditions of configurable analog hardware while preserving hardware simplicity; (ii) its demonstration through automatic foreground DC-offset calibration of a complete low-cost geophone conditioning chain; and (iii) an experimental evaluation on two independently assembled acquisition boards under multiple gain configurations using a single controller configuration across all reported conditions.

The remainder of this paper is organized as follows. Section II presents the proposed architecture. Section III describes the foreground calibration method. Section IV reports the experimental validation, followed by the discussion and conclusion.

II. Digitally Assisted Analog Front-End Architecture

The proposed Digitally Assisted Analog Front-End architecture combines a conventional analog acquisition path with temporary embedded digital assistance during foreground calibration. Unlike post-acquisition digital compensation, the embedded digital resources are employed only to establish the desired operating point of each analog stage. Once calibration is completed, the digital assistance is disabled and signal acquisition proceeds entirely through the calibrated analog signal path.

The architecture is implemented on a PSoC 5LP programmable mixed-signal platform. During foreground calibration, the embedded analog multiplexer, delta-sigma ADC, programmable digital filter, voltage DACs (VDACs), and embedded processor are temporarily repurposed as a mixed-signal feedback system that assists the analog circuitry while leaving the acquisition signal path unchanged. During normal operation, these resources resume their conventional acquisition functions.

The analog signal path, derived from the low-frequency geophone front end proposed by Ma *et al.* [15], is shown in Fig. 1. The conditioning chain consists of a programmable-gain instrumentation amplifier (PGA), a first-order active high-pass filter, an inverting summing stage, and a second-order low-pass filter. Four independently controlled VDACs shift the operating point of each stage during foreground calibration.

The embedded digital resources sequentially measure the output of each analog stage through the internal analog multiplexer and delta-sigma ADC. The filtered DC estimate determines the VDAC correction required to recenter each stage. Calibration proceeds from the PGA to the low-pass filter because each stage inherits the residual offset propagated from the preceding stages.

Assuming quasi-static offsets during calibration, negligible loading between stages, and linear superposition of DC contributions, the residual offset at the output of stage i can be expressed as

$$y_i^{\mathrm{DC}} = \sum_{j<i} A_{ij} y_j^{\mathrm{DC}} + b_i + K_i u_i, \tag{1}$$

where A_{ij} represents the DC transfer gain from stage j to stage i, b_i denotes the intrinsic offset generated by stage i, and K_i is the DC gain relating the injected VDAC voltage u_i to the output of the corresponding stage.

This sequential operating principle is the key characteristic of the proposed Digitally Assisted Analog Front-End architecture. Each stage is temporarily assisted until its operating point converges to the desired value. After foreground calibration, the digital assistance is disabled and the analog front end operates as a conventional feed-forward signal chain throughout data acquisition.

III. Foreground Calibration Method

As introduced in Section II, the proposed Digitally Assisted Analog Front-End architecture performs foreground calibration before data acquisition. During this stage, embedded digital resources temporarily assist each analog stage to establish its operating point. Once calibration is completed, the digital assistance is disabled and the analog signal chain resumes conventional feed-forward operation.

A. Calibration Workflow

The calibration workflow is illustrated in Fig. 2. At boot, after a gain change, or on command, the firmware loads the stored DAC codes and verifies whether the previously established calibration remains valid. Since DC estimation assumes a stationary geophone and slowly varying internal states, a valid operating point immediately returns the system to acquisition. Otherwise, the PGA, BP, SUM, and LP stages are calibrated sequentially with filter reset and settling after each stage. A watchdog timer and sample-count limits bound the process; during acquisition, the controller remains inactive, the LP stage is selected, and the VDAC codes remain constant.

During calibration, the programmable digital filter operates as a 128-tap FIR DC estimator, whereas during acquisition it is reconfigured with coefficients matched to the geophone bandwidth. After calibration, the acquisition coefficients, filter state, and routing are restored. A gain-indexed EEPROM record stores the valid-stage mask, four DAC codes, version, and cyclic-redundancy check (CRC); failed calibrations retain their bounded best codes without overwriting a valid record.

B. Control Law and Calibration Parameters

The same control framework is independently applied to each analog stage. For stage i, $\hat{y}_i[n]$ is the filtered signed ADC count, r_i is the desired reference count, and $e_i[n]$ is the residual. Since the calibration target is the reference level, $r_i = 0$ and therefore $e_i[n] = \hat{y}_i[n]$. The residual is first mapped to the 8-bit DAC domain:

$$\begin{aligned}\varepsilon_i[n] &= \mathrm{round}\left(e_i[n]\frac{q_{\mathrm{ADC}}}{q_{\mathrm{DAC}}}\right)\\ &= \mathrm{round}\left(\frac{e_i[n]V_{\mathrm{ADC,span}}N_{\mathrm{DAC}}}{N_{\mathrm{ADC}}V_{\mathrm{DAC,span}}}\right),\end{aligned} \tag{2}$$

where $q_x = V_{x,\mathrm{span}}/N_x$ is the quantization step and N_x the number of quantization levels. Nominal datasheet values are used throughout: $N_{\mathrm{ADC}} = 2^{18}$, $N_{\mathrm{DAC}} = 2^8$, $V_{\mathrm{ADC,span}} = 5000$ mV, and $V_{\mathrm{DAC,span}} = 4096$ mV, giving $q_{\mathrm{ADC}} = 5000/2^{18}$ mV/count and $q_{\mathrm{DAC}} = 16$ mV/code [9], [16].

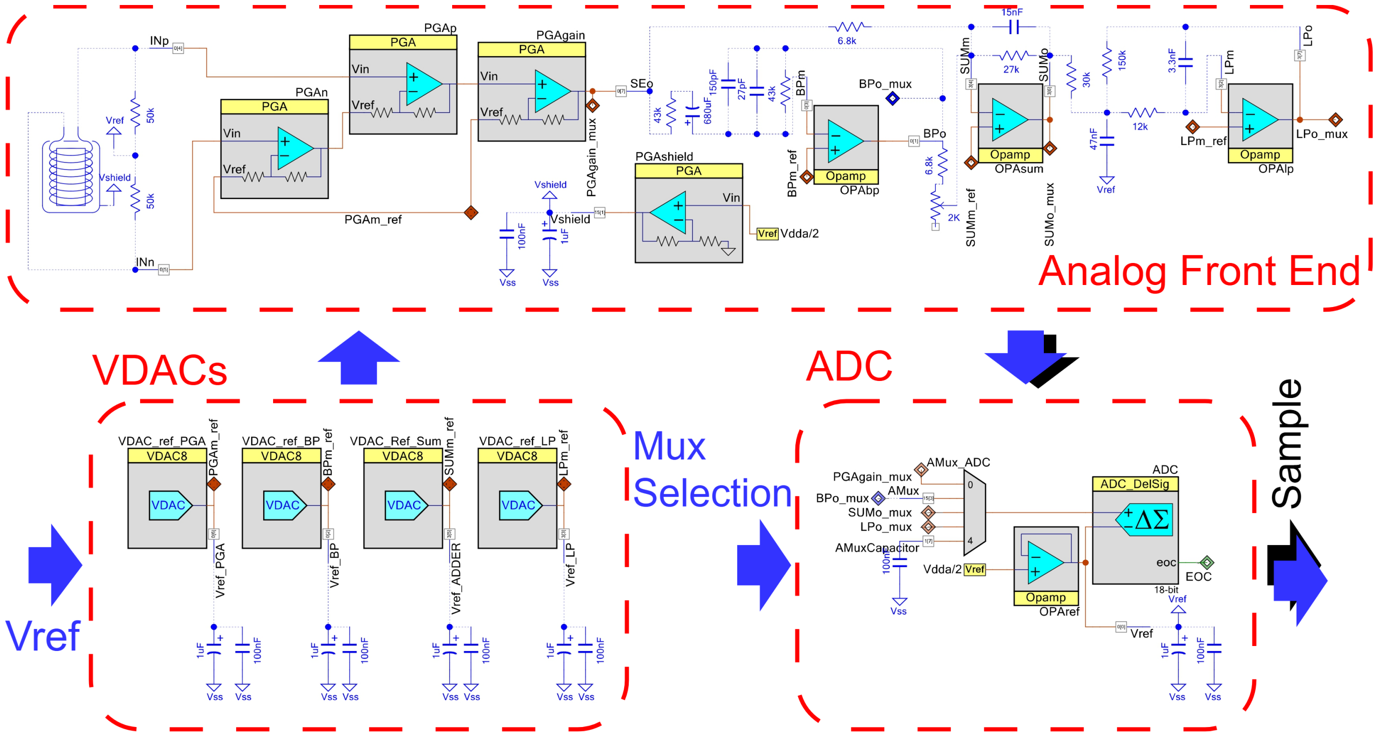

Fig. 1: Analog signal chain: four independent VDAC references trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

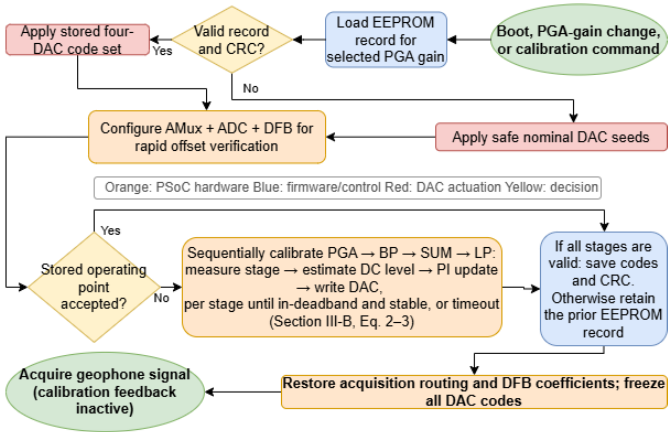

Fig. 2: Foreground calibration flow: stored-code verification, sequential trimming, and acquisition.

Equation 2 produces an integer error in equivalent DAC codes. To avoid oscillation between adjacent codes when the ideal operating point is not realizable, the implementation adopts the conservative one-step floor

$$\Delta_{i,\min}^{\text{impl}} = \max(1, \lceil |K_i| \rceil), \quad (3) \qquad \bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \le \Delta_i, \\ \varepsilon_i[n], & \text{otherwise}, \end{cases} \quad (4)$$

Inside the deadband, Eq. 4 suppresses integration, while lock and refinement determine convergence. Table I lists the adopted values satisfying $\Delta_i \ge \Delta_{i,\min}^{\text{impl}}$.

Under the assumptions of Section II, the architecture requires a measurable stage output together with sufficient trim authority and range. The controller is implemented in

TABLE I: Parameters used for the reported measurements. Deadbands are in equivalent DAC codes; S/L/R are settle/lock/refine samples at 2604 Hz, whereas Timeout is in seconds.

Stage	$\lvert K_i \rvert$	$\Delta_{i,\min}^{\text{impl}}$	Δ_i	$k_{\text{P},i}$	$k_{\text{I},i}$	S/L/R	Timeout (s)
PGA	1	1	18	10^{-3}	3×10^{-4}	512/512/1024	11.5
BP	1	1	3	10^{-4}	5×10^{-4}	512/1024/1024	17.3
SUM	7.89	8	10	10^{-4}	5×10^{-4}	512/512/1024	11.5
LP	6	6	8	10^{-4}	5×10^{-4}	1024/1024/2048	17.3

positional proportional-integral (PI) form,

$$c_i^*[n] = c_{i,\text{base}} - s_i \left[\frac{k_{\text{P},i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{\text{I},i}}{|K_i|} \sum_{q=0}^{n-1} \bar{\varepsilon}_i[q] \right], \quad (5)$$

where $c_{i,\text{base}}$ is the baseline feedforward code and $s_i = \operatorname{sgn}(K_i)$. The integral term accumulates accepted residuals and resets inside the deadband. The VDAC-to-node gains $|K_i|$ follow from the stage topology; for the two-op-amp instrumentation amplifier PGA [17], $K_{\text{PGA}} = 1$.

The command is clipped to the configured 0–255 DAC range. At either limit, integration is suspended whenever the residual would drive the command farther into saturation, preventing windup. Convergence requires a fixed number of consecutive in-deadband samples, while an adjacent-code trial is retained only if it reduces the filtered residual. At timeout, the best code is preserved and the stage is flagged.

The parameter set in Table I was selected empirically during preliminary bench tests and subsequently kept fixed

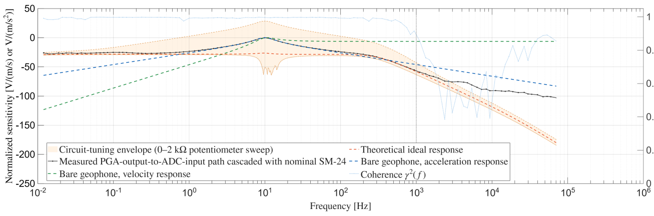

Fig. 3: Initial analog-path check: sinusoidal sweeps measured the PGA-output-to-ADC-input path, cascaded with the nominal SM-24 model (markers). Left: normalized magnitude, with the theoretical models and 0–2 kΩ tuning envelope as shape references. Right: local coherence. The dash-dotted line marks the rounded 1-kHz coherence onset. No calibrated seismic source available.

for all reported measurements. No systematic optimization was performed, demonstrating robust operation with a single controller configuration across all evaluated boards and gain settings. Section V discusses the selected value of $\Delta_{\mathrm{PGA}} = 18$. Finer trim resolution could be achieved through current-output DACs or buffered injection topologies [16] without modifying the foreground calibration method.

IV. Experimental Validation

The proposed Digitally Assisted Analog Front-End architecture was experimentally evaluated to verify that embedded digital assistance improves the effective operating conditions of the analog signal chain while preserving a conventional acquisition path during normal operation. The evaluation comprised analog front-end characterization, foreground calibration on two independently assembled geophone acquisition boards under multiple gain configurations, quantitative residual DC-offset measurements, and a field demonstration.

A. Analog Front-End Characterization

Prior to evaluating the proposed architecture, the frequency response of the analog signal chain was characterized independently of the geophone. An Agilent 33220A waveform generator excited the front end using overlapping logarithmic sweeps from 0.01 Hz to 200 kHz, while a Tektronix TDS 2004B [18] recorded the stage inputs and outputs. The measured PGA-output-to-ADC-input response was defined by the LP-output/PGA-output ratio (CH4/CH1).

For the approximate sensor-to-ADC response shown in Fig. 3, the measured electronics response was cascaded with the nominal SM-24 acceleration model [8]. All magnitude curves were normalized to facilitate comparison. Local coherence first falls below 0.9 near 0.84 kHz, indicating that the increasing disagreement above 1 kHz is measurement-limited rather than a verified change in circuit behavior. This experiment therefore provides a consistency check of the analog front end rather than an end-to-end validation of the sensing chain.

B. Prototype and Comparison Conditions

The experimental validation comprised eight board–gain configurations: all five PGA settings on Board A together with the $\times16$, $\times32$, and $\times50$ settings on Board B. Each configuration was evaluated through ten paired restarts ($N = 10$), representing repeated measurements of the same board and gain rather than independent devices.

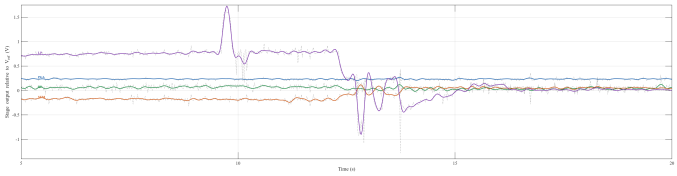

Fig. 4: Stage deviations relative to V_{ref} during calibration at $\times16$ (total differential gain $\times32$). Dashed light-gray traces are oscilloscope measurements; colored curves are resampled 128-tap-FIR reconstructions, not internal logs (Section V).

A single fixed-reference baseline, $\mathbf{c}_{\mathrm{base}}$, selected empirically on Board B at $\times50$, was applied unchanged to all reported configurations. For each calibrated condition, the four analog stages were processed sequentially and the resulting DAC codes remained fixed during acquisition. Post-calibration EEPROM writes were disabled so that every restart began from the same baseline.

Board B at $\times1$ and $\times4$ were excluded because their baseline residuals already lay within the calibration deadbands, preventing corrective updates. A separate narrower-deadband diagnostic confirmed the expected behavior (Section V).

C. Residual DC-Offset Measurement and Results

The LP output was selected as the primary performance metric because it directly feeds the acquisition ADC. Measurements at the PGA, BP, and SUM outputs document the sequential propagation of the foreground calibration through the analog signal chain.

A four-channel Tektronix TDS 2004B recorded the mean output voltage of the PGA, BP, SUM, and LP stages over a 1-s window after settling. Probe grounds were connected to $V_{\mathrm{ref}} = V_{\mathrm{DDA}}/2$, so each channel measured $V_{\mathrm{stage}} - V_{\mathrm{ref}}$. Within each restart, the fixed-reference condition was always recorded before the calibrated condition, preserving paired measurements.

Table II reports the mean and sample standard deviation of the normalized residual offset over ten paired restarts. Figure 4 shows representative calibration trajectories reconstructed by applying the 128-tap FIR used by the controller to oscilloscope records resampled at the controller rate.

Across the eight reported board–gain configurations, the proposed Digitally Assisted Analog Front-End consistently reduced the LP residual DC offset without requiring board-specific or gain-specific controller retuning. Relative to the shared fixed-reference baseline, the residual offset decreased from 2.9–45.8%FS to 0.4–3.9%FS. Individual calibrated observations ranged from 0.1–5.2%FS (Table II). A one-sided sign test on the paired LP measurements rejected the null hypothesis of no reduction in all eight configurations (10 of 10 paired restarts, exact $p \approx 10^{-3}$).

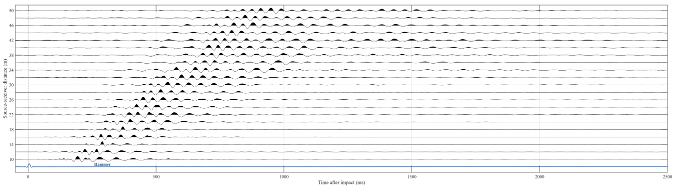

Fig. 5: Wiggle-trace field record acquired after foreground calibration; the trigger and first-arrival moveout demonstrate multichannel operation over the 10–50 m line.

D. Field Demonstration

A field experiment verified that foreground calibration integrates seamlessly with normal multichannel acquisition under representative operating conditions. Figure 5 shows a 10–50 m seismic record acquired after foreground calibration, with the geophones remaining stationary during the calibration stage. The resolved first-arrival moveout confirms successful multichannel operation. The only timeout observed during laboratory and field measurements occurred when a board was powered while its geophone was moving, violating the quasi-static assumption required during foreground calibration.

V. Discussion and Conclusion

The proposed Digitally Assisted Analog Front-End architecture demonstrates that embedded mixed-signal resources can improve the effective operating conditions of configurable analog hardware without modifying the analog acquisition path during normal operation. Unlike digital post-processing, the embedded resources temporarily assist the analog circuitry during foreground calibration and are subsequently disengaged, allowing the calibrated signal chain to operate as a conventional feed-forward analog front end.

From an instrumentation perspective, the proposed architecture differs from two conventional approaches. Unlike precision analog front ends designed to operate close to the sensor's Johnson-noise limit [19], it trades analog component precision for embedded measurement and digital assistance. Likewise, unlike purely digital offset compensation, it physically recenters observable additive DC errors within the available VDAC authority and range instead of correcting them numerically after acquisition. The reported improvements therefore reflect the performance of the complete architecture using a common fixed-reference baseline and a single controller configuration across all evaluated board and gain combinations rather than per-stage parameter optimization.

The proposed foreground calibration strategy also differs from continuous DC-servo architectures, which inherently introduce a settling-versus-bandwidth tradeoff through permanently closed feedback loops [14]. Here, digital assistance is active only during foreground calibration. Once calibration is completed, the controller is disabled, the VDAC outputs remain constant, and the acquisition path contains no continuously active digital feedback.

The present implementation should be interpreted as a proof of architectural feasibility rather than as an optimized analog front end. The relatively modest PGA improvement illustrates this point. Because $|K_{\mathrm{PGA}}| = 1$, the selected deadband ($\Delta_{\mathrm{PGA}} = 18$) accepts integer-error buckets through approximately ± 288 mV (continuous edge near ± 296 mV after rounding). Consequently, the 0.24-V residual shown in Fig. 4 reflects the selected stopping criterion rather than controller instability. A separate diagnostic experiment on Board B at $\times 4$ gain, performed outside the reported comparison, reduced the LP residual from 2.98 to 0.84%FS using a narrower deadband, indicating that further improvement is possible through parameter optimization. Likewise, the only increase observed at the reported precision (Board A, SUM stage, $\times 50$ gain, from 0.4 to 0.8%FS) results from deadband acceptance and does not propagate to the downstream LP output.

The timing results also support the practical feasibility of foreground calibration. Visible correction in Fig. 4 spans approximately 7 s, while informal bench observations indicated completion in about 10 s. The configured timeout windows total 57.6 s (approximately 60.5 s including settling and refinement), excluding stored-code verification and switching overhead. No laboratory timeout occurred during the reported experiments. Nevertheless, the present study was intentionally limited to two independently assembled boards and fixed calibration ordering; consequently, it does not establish population-level statistical performance, long-term drift behavior, or optimal controller tuning.

Overall, the experimental results demonstrate that the proposed Digitally Assisted Analog Front-End architecture provides a practical means of improving the operating conditions of low-cost configurable analog front ends by temporarily exploiting the embedded mixed-signal resources already available in programmable systems-on-chip. Implemented on a low-cost geophone acquisition platform, the proposed foreground DC-offset calibration consistently reduced the residual offset at the analog front-end output using a single controller configuration across all reported boards and gain settings while preserving a conventional analog acquisition path. Although demonstrated through foreground DC-offset calibration, the proposed architecture is not restricted to this function. Foreground calibration represents one practical example of how embedded digital resources can temporarily assist configurable analog hardware before conventional data acquisition. Additional digitally assisted functions, including gain calibration, thermal compensation, and adaptive bias adjustment, will be investigated in future work.

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TABLE II: Residual DC offset (%FS of the nominal 5-V ADC span), mean $\pm$ sample standard deviation over N=10 paired baseline-seeded restarts per reported board/gain cell. Within each cell, the upper and lower values correspond to the shared fixed-reference and calibrated conditions, respectively.

Stage	A, $\times 1$	A, $\times 4$	A, $\times 16$	B, $\times 16$	A, $\times 32$	B, $\times 32$	A, $\times 50$	B, $\times 50$
PGA	3.5 ± 0.2 3.4 ± 0.3	2.4 ± 0.1 2.1 ± 0.3	2.4 ± 0.1 1.9 ± 0.2	2.1 ± 0.1 1.7 ± 0.1	1.0 ± 0.2 0.5 ± 0.2	1.0 ± 0.2 0.5 ± 0.2	1.2 ± 0.3 1.0 ± 0.5	2.5 ± 0.1 1.8 ± 0.3
BP	1.5 ± 2.2 1.2 ± 0.2	4.9 ± 0.3 1.0 ± 0.2	3.8 ± 0.9 1.5 ± 1.5	0.66 ± 0.02 0.66 ± 0.02	7.0 ± 0.4 1.0 ± 0.5	0.66 ± 0.01 0.66 ± 0.01	1.0 ± 0.7 0.9 ± 0.5	0.65 ± 0.01 0.65 ± 0.02
SUM	$\mathbf{21.8 \pm 1.1}^{\dagger}$ 0.8 ± 0.2	6.7 ± 1.0 1.0 ± 0.3	4.9 ± 1.5 1.4 ± 1.5	4.5 ± 0.8 0.6 ± 0.5	7.4 ± 0.3 0.6 ± 0.2	8.0 ± 0.8 1.2 ± 0.3	0.4 ± 0.2 $\mathbf{0.8 \pm 0.6}^{\ddagger}$	0.9 ± 0.6 0.9 ± 0.6
LP	$\mathbf{45.8 \pm 3.1}^{\dagger}$ 3.2 ± 0.9	$\mathbf{42.6 \pm 4.2}^{\dagger}$ 3.9 ± 0.8	$\mathbf{29.6 \pm 6.8}^{\dagger}$ 2.3 ± 1.4	3.0 ± 0.1 0.7 ± 0.1	$\mathbf{12.4 \pm 1.3}^{\dagger}$ 1.7 ± 0.7	2.9 ± 0.1 0.4 ± 0.2	4.1 ± 2.1 1.2 ± 0.4	2.9 ± 0.1 0.7 ± 0.2

Lower is better. †Fixed reference ≥ 10%FS. ‡The marked calibrated mean is larger than its baseline; equality after rounding is not marked (Section V).

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